

Project: Data-driven Insights using Python: FoodHub Data Analysis

Problem Statement

Context

The number of restaurants in New York is increasing day by day. Lots of students and busy professionals rely on those restaurants due to their hectic lifestyles. An online food delivery service is a great option for them. It provides them with good food from their favorite restaurants. A food aggregator company FoodHub, offers access to multiple restaurants through a single smartphone app. The app allows the restaurants to receive a direct online order from a customer. The app assigns a delivery person from the company to pick up the order after it is confirmed by the restaurant. The delivery person then uses the map to reach the restaurant and waits for the food package. Once the food package is handed over to the delivery person, he/she confirm the pick-up in the app and travel to the customer's location to deliver the food. The delivery person confirms the drop-off in the app after delivering the food package to the customer. The customer can rate the order in the app. The food aggregator earns money by collecting a fixed margin of the delivery order from the restaurants.

Objective

The food aggregator company has stored the data of the different orders made by the registered customers in its online portal. They want to analyze the data to get a fair idea about the demand for different restaurants, which will help them enhance their customer experience. Suppose you are hired as a Data Scientist in this company, and the Data Science team has shared some of the key questions that need to be answered. Perform the data analysis to find answers to these questions that will help the company improve its business.

Data Description

The data contains different data related to a food order. The detailed data dictionary is given below.

- * `order_id`: Unique ID of the order
- * `customer_id`: ID of the customer who ordered the food
- * `restaurant_name`: Name of the restaurant
- * `cuisine_type`: Cuisine ordered by the customer
- * `cost_of_the_order`: Cost of the order
- * `day_of_the_week`: Indicates whether the order is placed on a weekday or weekend (The weekday is from Monday to Friday and the weekend is Saturday and Sunday)
- * `rating`: Rating given by the customer out of 5
- * `food_preparation_time`: Time (in minutes) taken by the restaurant to prepare the food. This is calculated by taking the difference between the timestamps of the restaurant's order confirmation and the delivery person's pick-up confirmation.
- * `delivery_time`: Time (in minutes) taken by the delivery person to deliver the food package. This is calculated by taking the difference between the timestamps of the delivery person's pick-up confirmation and drop-off information

Let us start by importing the required libraries

```
In [ ]: # Installing the libraries with the specified version.
!pip install numpy==1.25.2 pandas==1.5.3 matplotlib==3.7.1 seaborn==0.13.1 -q
--user
```

Note:

- After running the above cell, kindly restart the runtime (for Google Colab) or notebook kernel (for Jupyter Notebook), and run all cells sequentially from the next cell.
- On executing the above line of code, you might see a warning regarding package dependencies. This error message can be ignored as the above code ensures that all necessary libraries and their dependencies are maintained to successfully execute the code in **this notebook**.

```
In [1]: # Import libraries for data manipulation
import numpy as np
import pandas as pd

# Import libraries for data visualization
import matplotlib.pyplot as plt
import seaborn as sns
```

Understanding the structure of the data

```
In [2]: # uncomment and run the below code snippets if the dataset is present in the
        # Google Drive
        from google.colab import drive
        drive.mount('/content/drive')
```

Mounted at /content/drive

```
In [3]: # Write your code here to read the data
        df = pd.read_csv('/content/drive/My Drive/Colab Notebooks/foodhub_order.csv')
```

```
In [ ]: # Write your code here to view the first 5 rows
        df.head(5)
```

	order_id	customer_id	restaurant_name	cuisine_type	cost_of_the_order	c
0	1477147	337525	Hangawi	Korean	30.75	V
1	1477685	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	V
2	1477070	66393	Cafe Habana	Mexican	12.23	V
3	1477334	106968	Blue Ribbon Fried Chicken	American	29.20	V
4	1478249	76942	Dirty Bird to Go	American	11.59	V

Question 1: How many rows and columns are present in the data? [0.5 mark]

```
In [4]: # Write your code here
        df.shape
```

Out[4]: (1898, 9)

Observations: The data has 1898 rows and 9 columns

Question 2: What are the datatypes of the different columns in the dataset? (The info() function can be used) [0.5 mark]

```
In [5]: # Write your code here
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1898 entries, 0 to 1897
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   order_id                             1898 non-null   int64
1   customer_id                         1898 non-null   int64
2   restaurant_name                     1898 non-null   object
3   cuisine_type                       1898 non-null   object
4   cost_of_the_order                   1898 non-null   float64
5   day_of_the_week                    1898 non-null   object
6   rating                             1898 non-null   object
7   food_preparation_time              1898 non-null   int64
8   delivery_time                      1898 non-null   int64
dtypes: float64(1), int64(4), object(4)
memory usage: 133.6+ KB
```

Observations: All the columns seem to have the correct datatype, would consider changing the 'day_of_the_week' column type to datetime format but no need since the column is only displaying a string no a year month date format

Question 3: Are there any missing values in the data? If yes, treat them using an appropriate method. [1 mark]

```
In [6]: # Write your code here
df.isnull().sum()
```

```
Out[6]:
```

	0
order_id	0
customer_id	0
restaurant_name	0
cuisine_type	0
cost_of_the_order	0
day_of_the_week	0

	0
rating	0
food_preparation_time	0
delivery_time	0

dtype: int64

Observations: There is no missing data which is nice. Checked the ratings to make sure but if no rating was left, it just has the string 'not given'.

Question 4: Check the statistical summary of the data. What is the minimum, average, and maximum time it takes for food to be prepared once an order is placed? [2 marks]

In [7]:

```
# Write your code here
df.describe()
```

Out[7]:

	order_id	customer_id	cost_of_the_order	food_preparation_time
count	1.898000e+03	1898.000000	1898.000000	1898.000000
mean	1.477496e+06	171168.478398	16.498851	27.371970
std	5.480497e+02	113698.139743	7.483812	4.632481
min	1.476547e+06	1311.000000	4.470000	20.000000
25%	1.477021e+06	77787.750000	12.080000	23.000000
50%	1.477496e+06	128600.000000	14.140000	27.000000
75%	1.477970e+06	270525.000000	22.297500	31.000000
max	1.478444e+06	405334.000000	35.410000	35.000000

Observations: Seems the avg time, min time, and max time for food preparation is 27 min, 20 min, and 35 min respectively and then delivery times are similar .

Question 5: How many orders are not rated? [1 mark]

```
In [8]: df[df['rating'] == 'Not given'].shape[0]
```

```
Out[8]: 736
```

Observations: Out of the 1898 ratings, 736 of them have no rating or have 'Not given'

Exploratory Data Analysis (EDA)

Univariate Analysis

Question 6: Explore all the variables and provide observations on their distributions. (Generally, histograms, boxplots, countplots, etc. are used for univariate exploration.) [9 marks]

```
In [9]: # Dropping order id cause not important
df.drop('order_id', axis = 1, inplace =True)
```

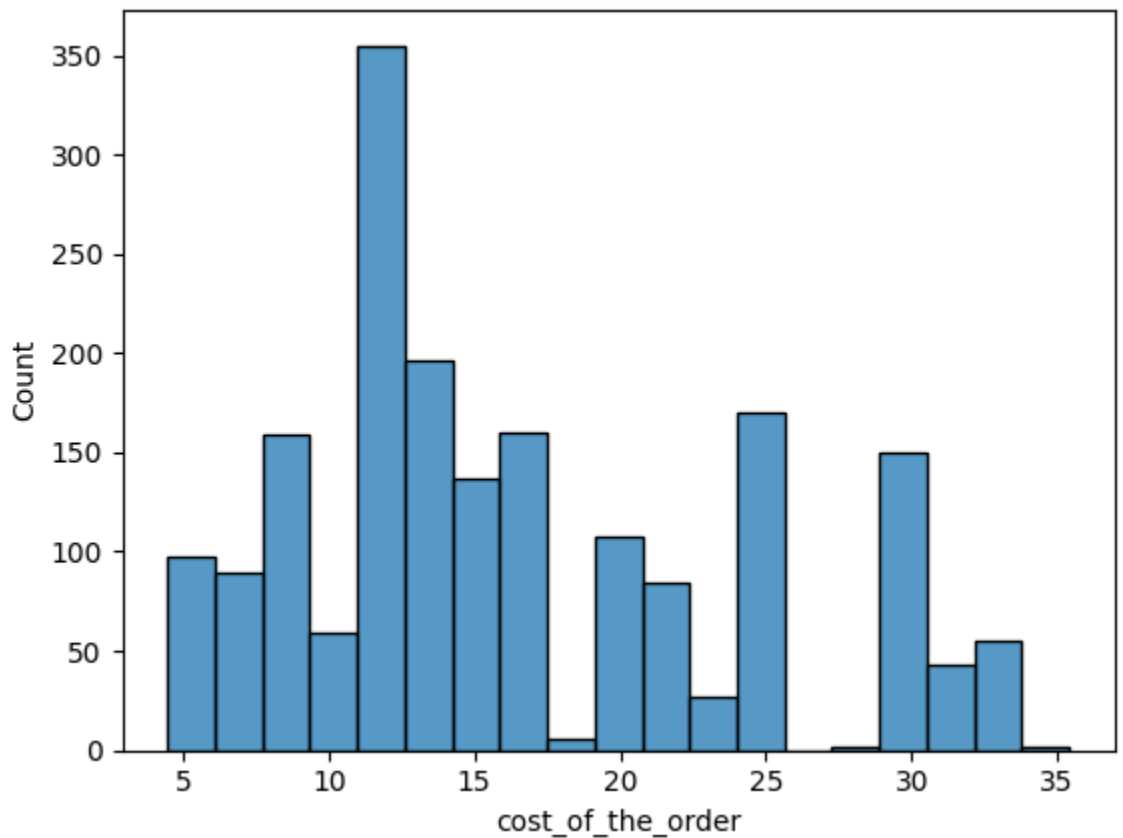
```
In [10]: #Making sure columns are not in the new dataframe
df.head(5)
```

```
Out[10]:
```

	customer_id	restaurant_name	cuisine_type	cost_of_the_order	day_of_the_
0	337525	Hangawi	Korean	30.75	Weekend
1	358141	Blue Ribbon Sushi Izakaya	Japanese	12.08	Weekend
2	66393	Cafe Habana	Mexican	12.23	Weekday
3	106968	Blue Ribbon Fried Chicken	American	29.20	Weekend
4	76942	Dirty Bird to Go	American	11.59	Weekday

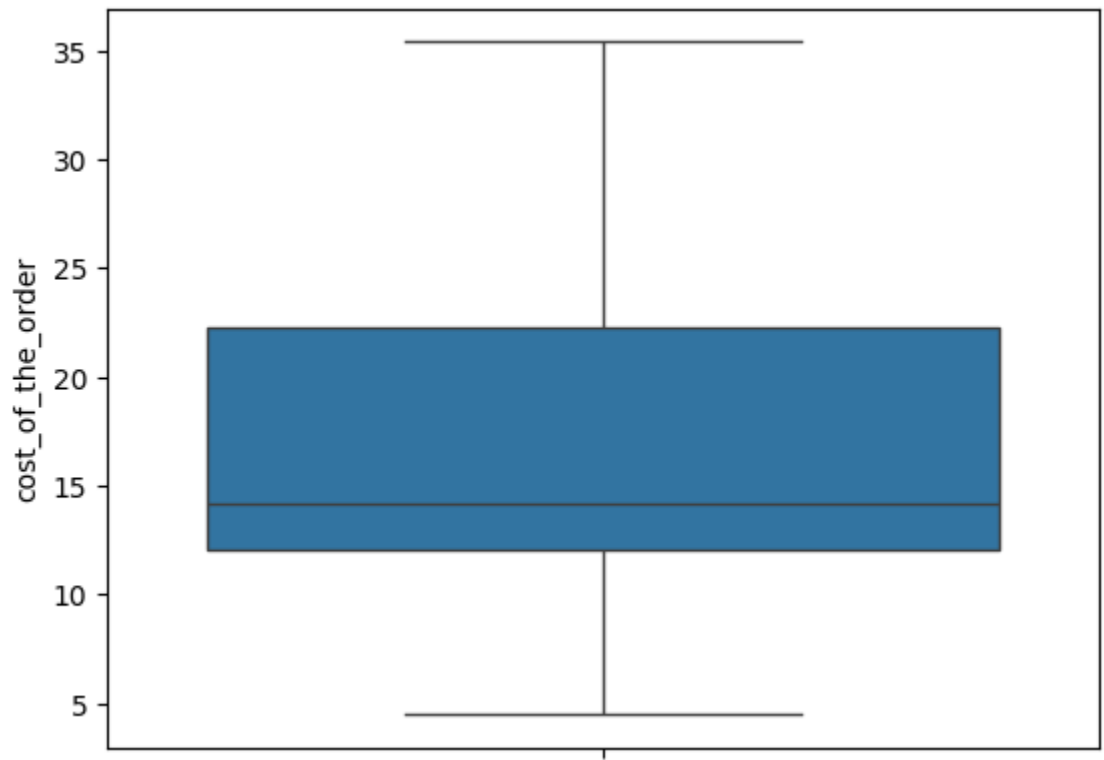
```
In [11]: # The plot is slightly right skewed
# Around 350 of the orders were $12-$13 but
sns.histplot(df['cost_of_the_order'])
```

Out[11]: <Axes: xlabel='cost_of_the_order', ylabel='Count'>



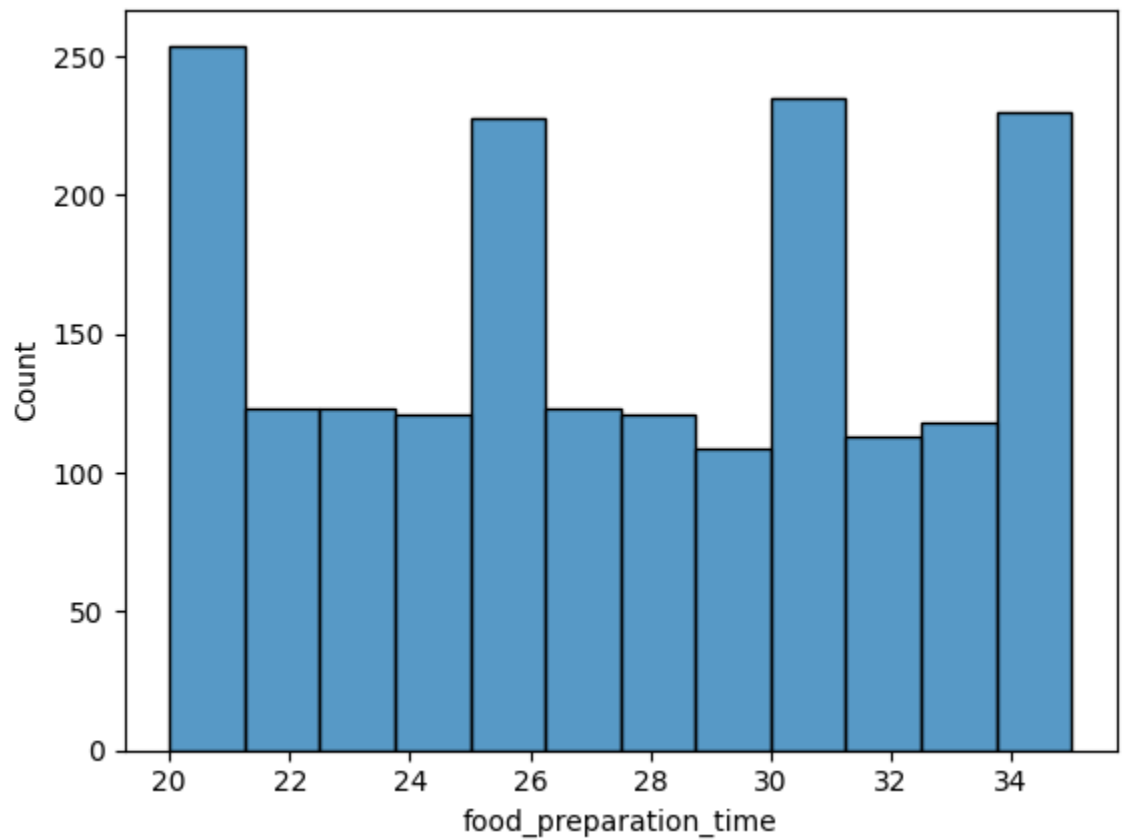
```
In [12]: #Checking box plot of variables to see if anything new comes up
sns.boxplot(df['cost_of_the_order'])
```

Out[12]: <Axes: ylabel='cost_of_the_order'>



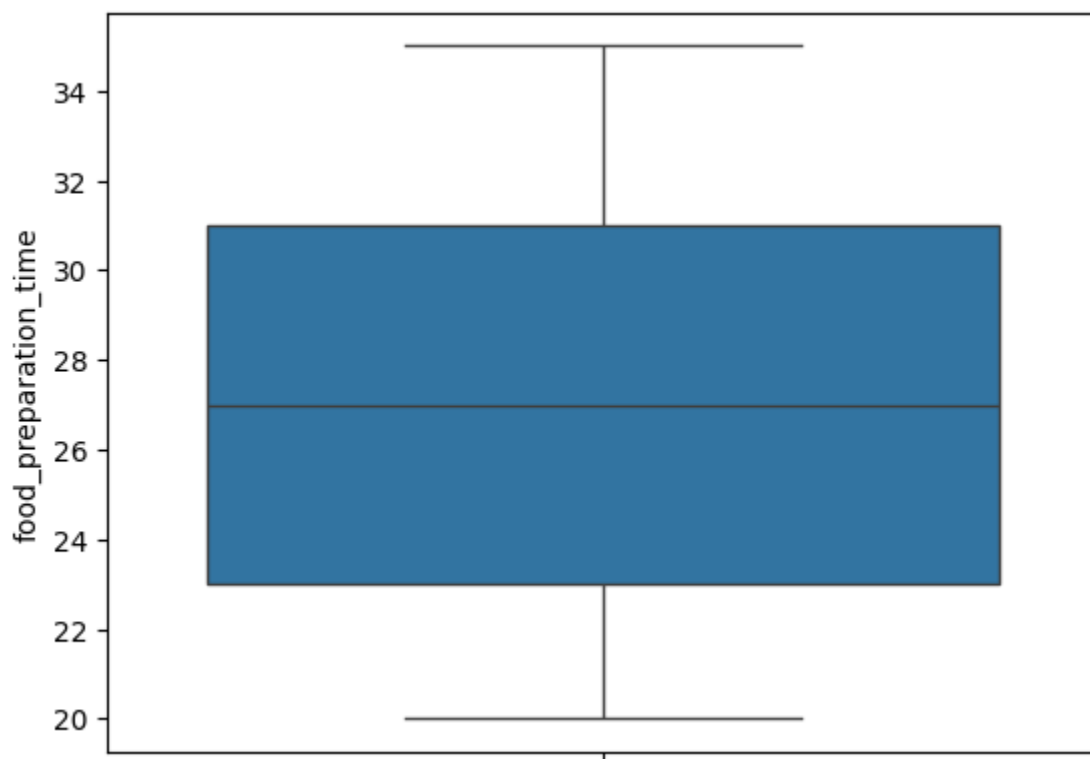
```
In [13]: sns.histplot(df['food_preparation_time'])
```

```
Out[13]: <Axes: xlabel='food_preparation_time', ylabel='Count'>
```



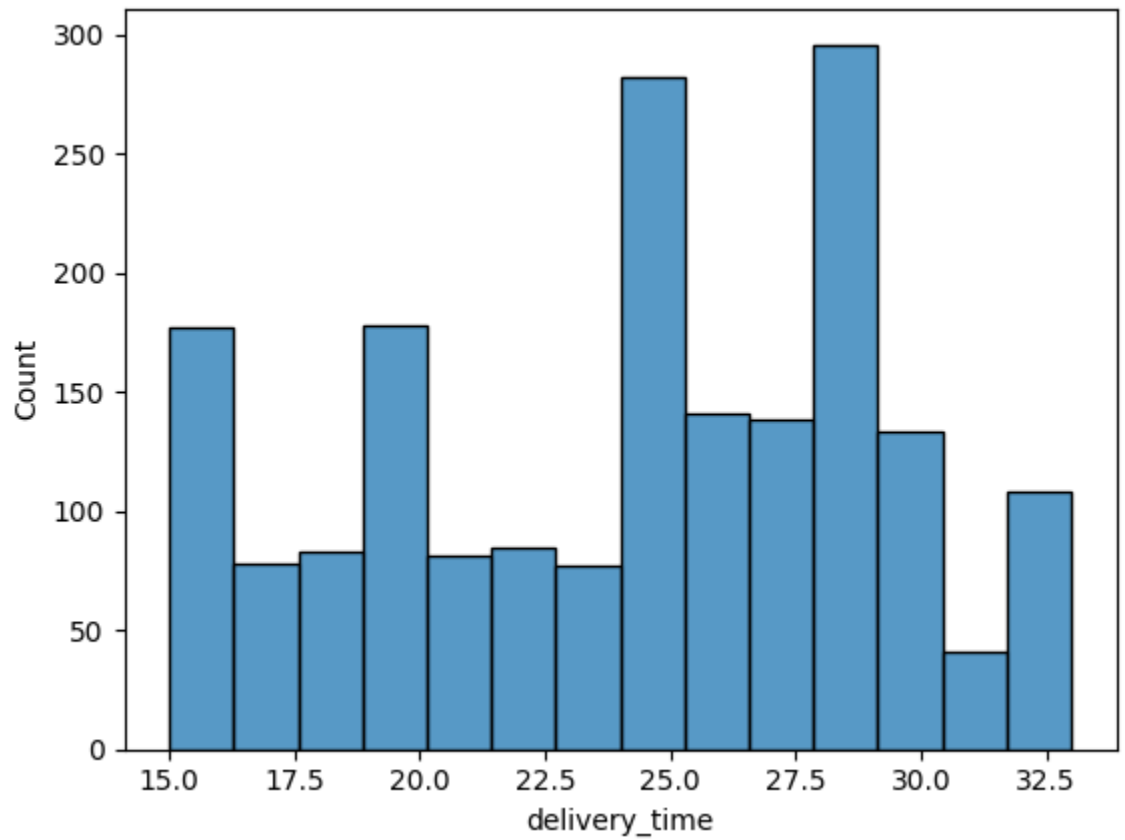

```
In [14]: sns.boxplot(df['food_preparation_time'])
# Median of food prep time is around 26 min
```

Out[14]: <Axes: ylabel='food_preparation_time'>



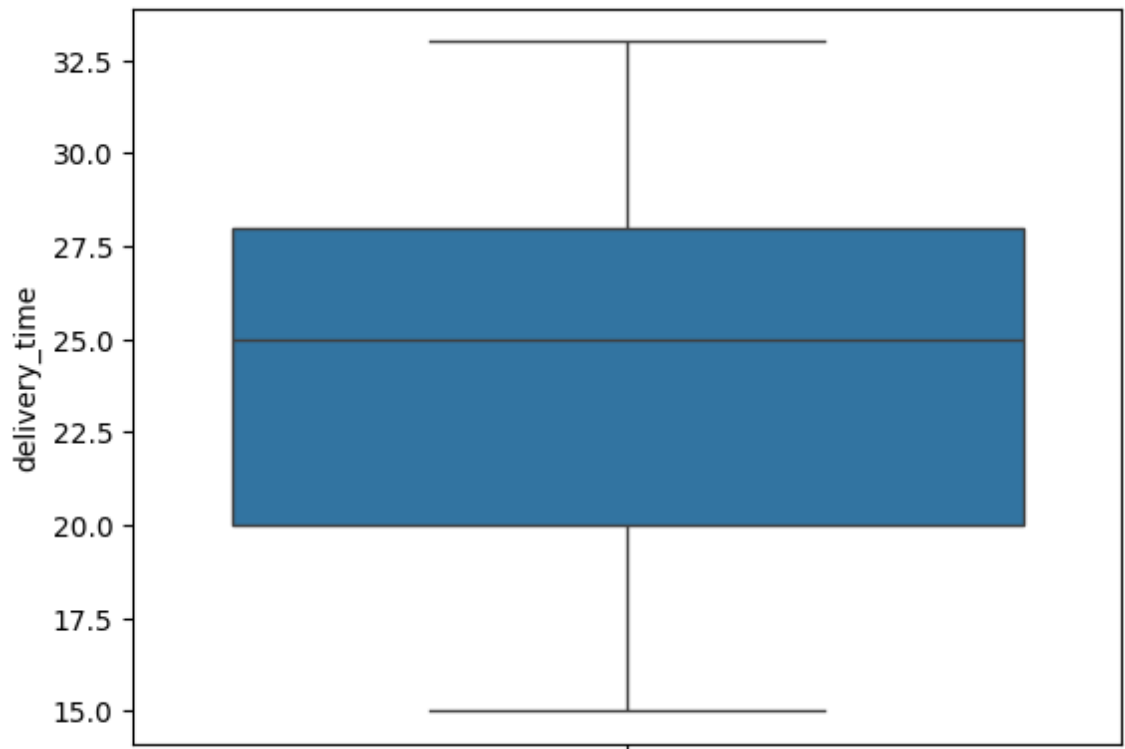
```
In [15]: sns.histplot(df['delivery_time'])
```

Out[15]: <Axes: xlabel='delivery_time', ylabel='Count'>

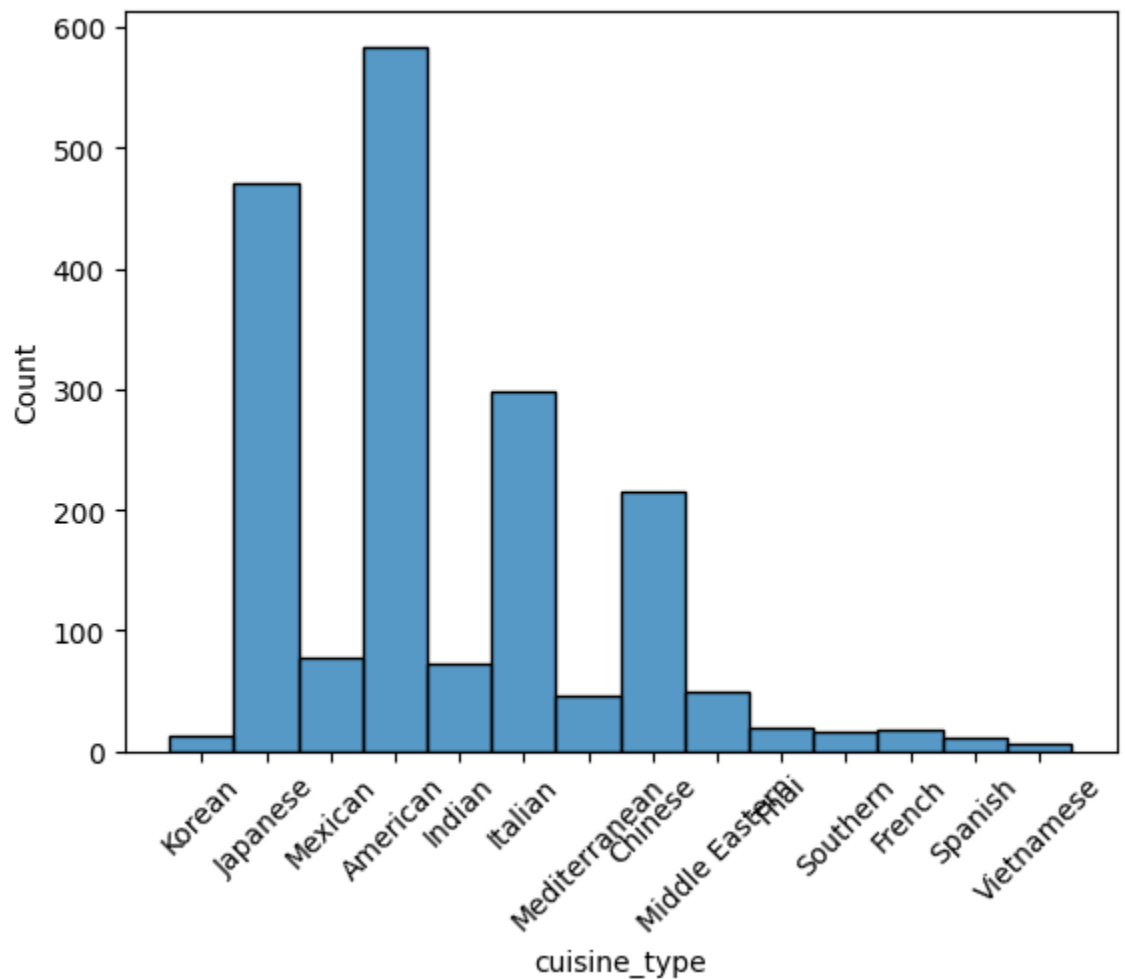


```
In [16]: sns.boxplot(df['delivery_time'])  
#Seems median of delivery time is around 25 min
```

```
Out[16]: <Axes: ylabel='delivery_time'>
```

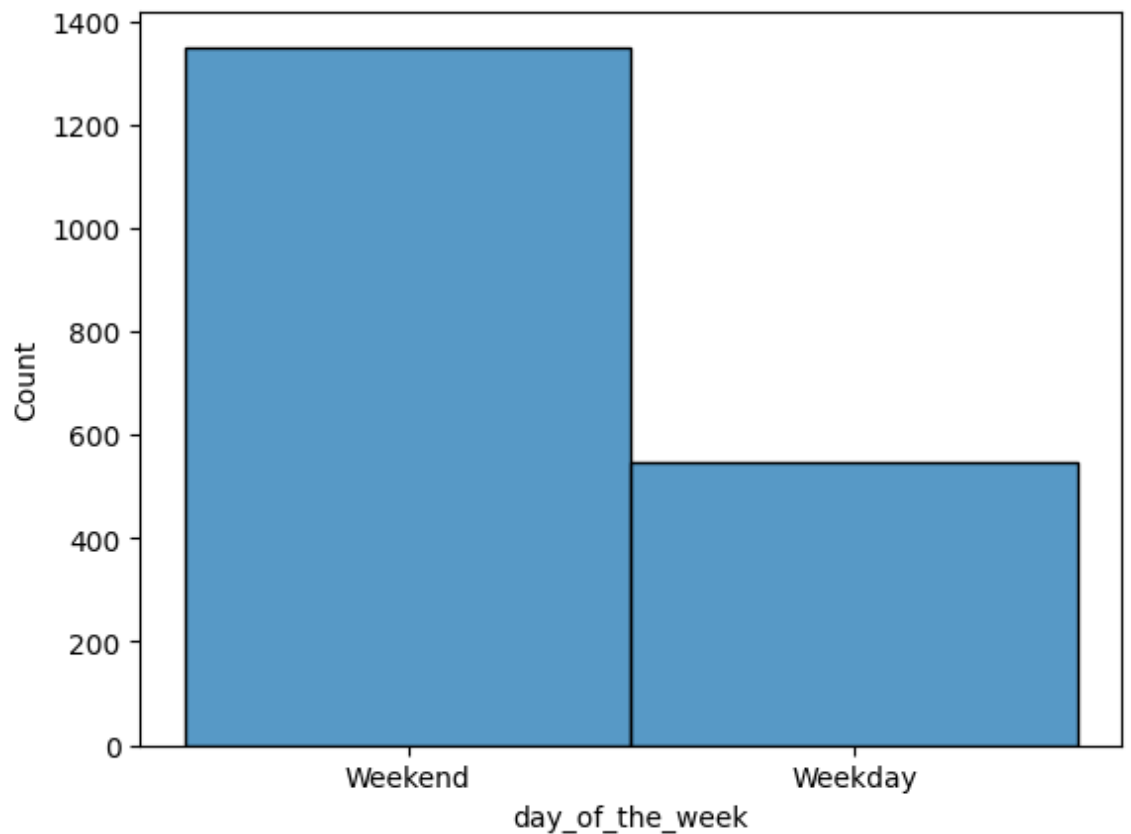


```
In [17]: sns.histplot(df['cuisine_type'])
plt.xticks(rotation=45)
plt.show()
#Most popular cuisines seem to be japanese, american, italian, and mediterranean
#While all other cuisines are ordered less than 100 times
```



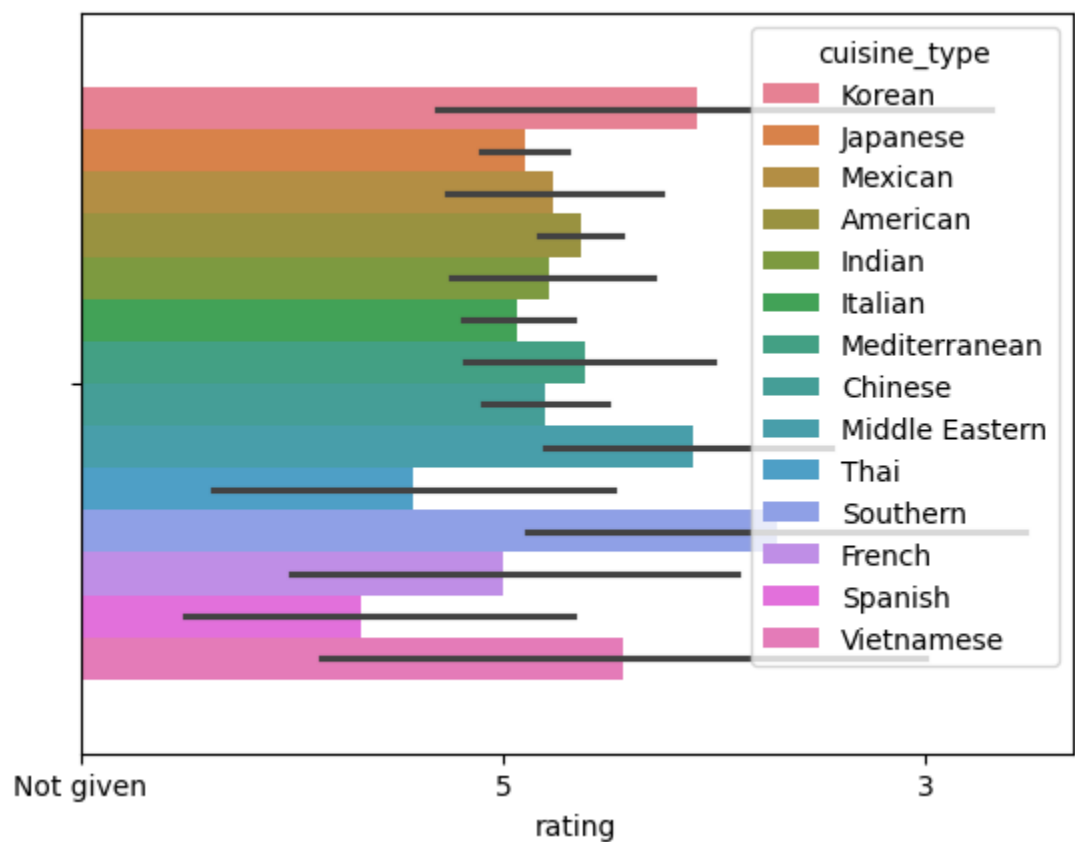
```
In [18]: sns.histplot(df['day_of_the_week'])
# More people order on the weekends compared to the weekdays
```

Out[18]: <Axes: xlabel='day_of_the_week', ylabel='Count'>



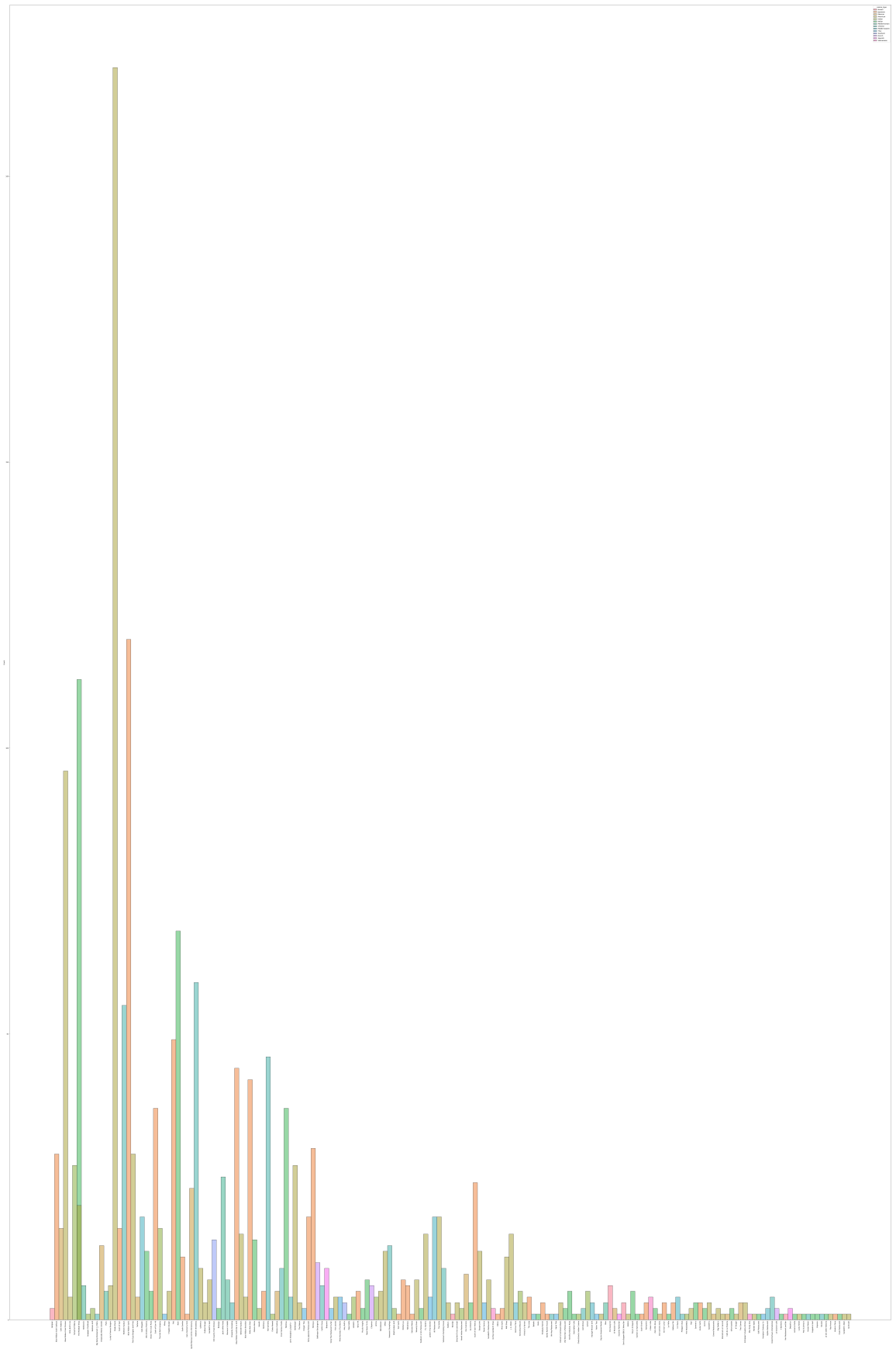
```
In [19]: sns.barplot(data = df, x = 'rating', hue= 'cuisine_type')
```

```
Out[19]: <Axes: xlabel='rating'>
```



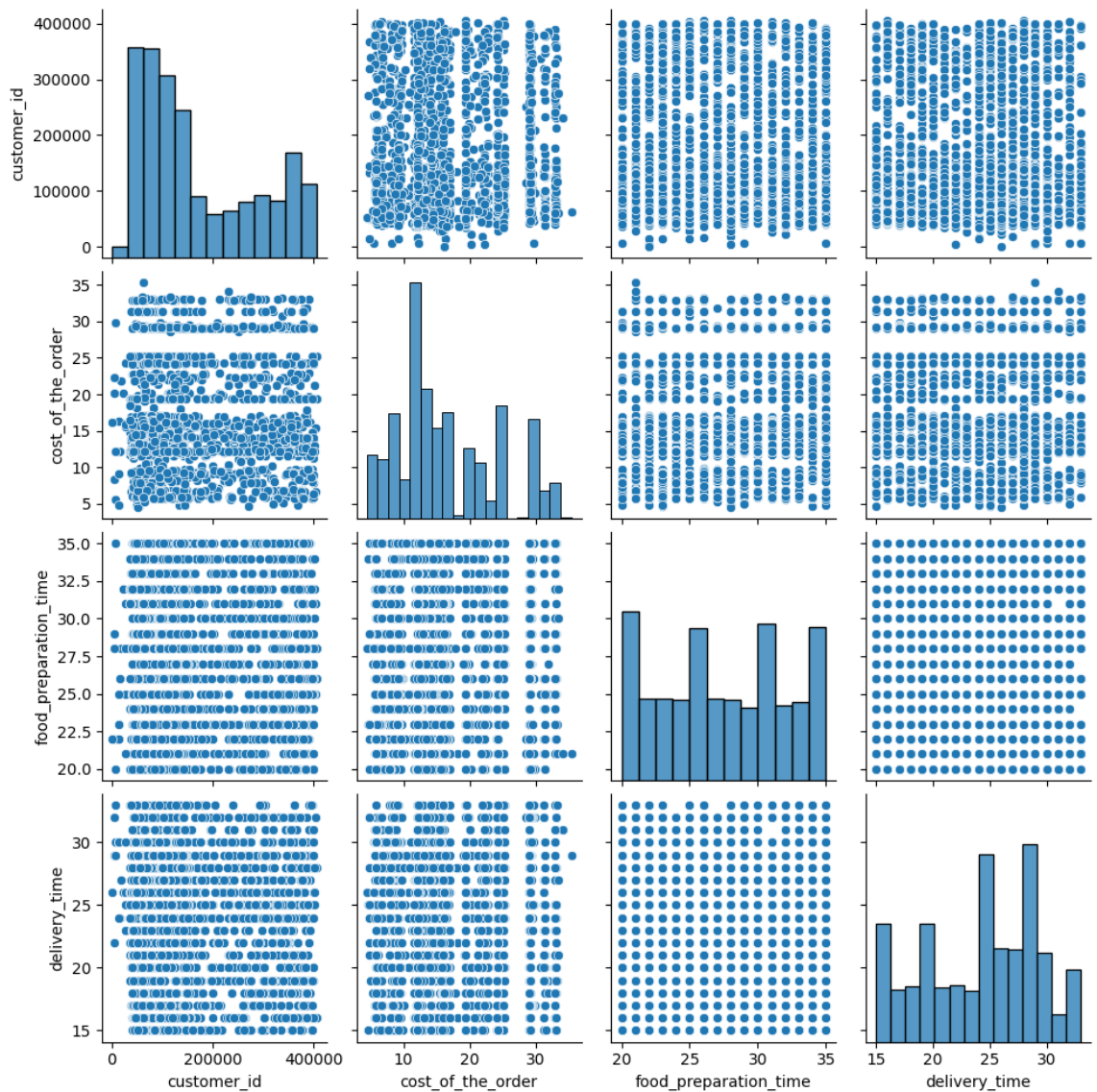
```
In [20]: plt.figure(figsize=(100,150))
sns.histplot(data = df, x= 'restaurant_name', hue= 'cuisine_type')
plt.xticks(rotation=90)
plt.show()
# Shake shack is the most popular resturant
# Blue ribbon has two places in the top 5 most popular resturants
```

```
/usr/local/lib/python3.12/dist-
packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 140 (\x8c)
missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
/usr/local/lib/python3.12/dist-
packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 142 (\x8e)
missing from font(s) DejaVu Sans.
    fig.canvas.print_figure(bytes_io, **kw)
```



```
In [21]: sns.pairplot(df)
# All the scatterplots are constant
```

```
Out[21]: <seaborn.axisgrid.PairGrid at 0x7b83799af530>
```



Question 7: Which are the top 5 restaurants in terms of the number of orders received? [1 mark]

```
In [22]: # Write the code here
df['restaurant_name'].value_counts().head(5)
```

Out[22]:

restaurant_name	count
Shake Shack	219
The Meatball Shop	132
Blue Ribbon Sushi	119
Blue Ribbon Fried Chicken	96

	count
restaurant_name	
Parm	68

dtype: int64

Observations: Shake shack is the most popular restaurant

Question 8: Which is the most popular cuisine on weekends? [1 mark]

```
In [23]: # Write the code here
df[df['day_of_the_week'] == 'Weekend']['cuisine_type'].value_counts().head(1)
```

Out[23]:

	count
cuisine_type	
American	415

dtype: int64

Observations: American is the most popular cuisine on the weekends

Question 9: What percentage of the orders cost more than 20 dollars? [2 marks]

```
In [24]: # Write the code here
df[df['cost_of_the_order'] > 20].shape[0] / df.shape[0]
```

Out[24]: 0.2924130663856691

Observations: Around 29% of orders are more than 20 dollars

Question 10: What is the mean order delivery time? [1 mark]


```
In [25]: # Write the code here
df['delivery_time'].mean()
```

```
Out[25]: np.float64(24.161749209694417)
```

```
#### Observations: The mean of the delivery time column is 24.16 min
```

Question 11: The company has decided to give 20% discount vouchers to the top 5 most frequent customers. Find the IDs of these customers and the number of orders they placed. [1 mark]

```
In [26]: # Write the code here
df['customer_id'].value_counts().head(5)
```

```
Out[26]:
```

	count
customer_id	
52832	13
47440	10
83287	9
250494	8
259341	7

```
dtype: int64
```

```
#### Observations: The highest number of orders a customer places was 13 orders
```

Multivariate Analysis

Question 12: Perform a multivariate analysis to explore relationships between the important variables in the dataset and provide observations. (It is a good idea to explore relations between numerical variables as well as relations between numerical and categorical variables) [10 marks]

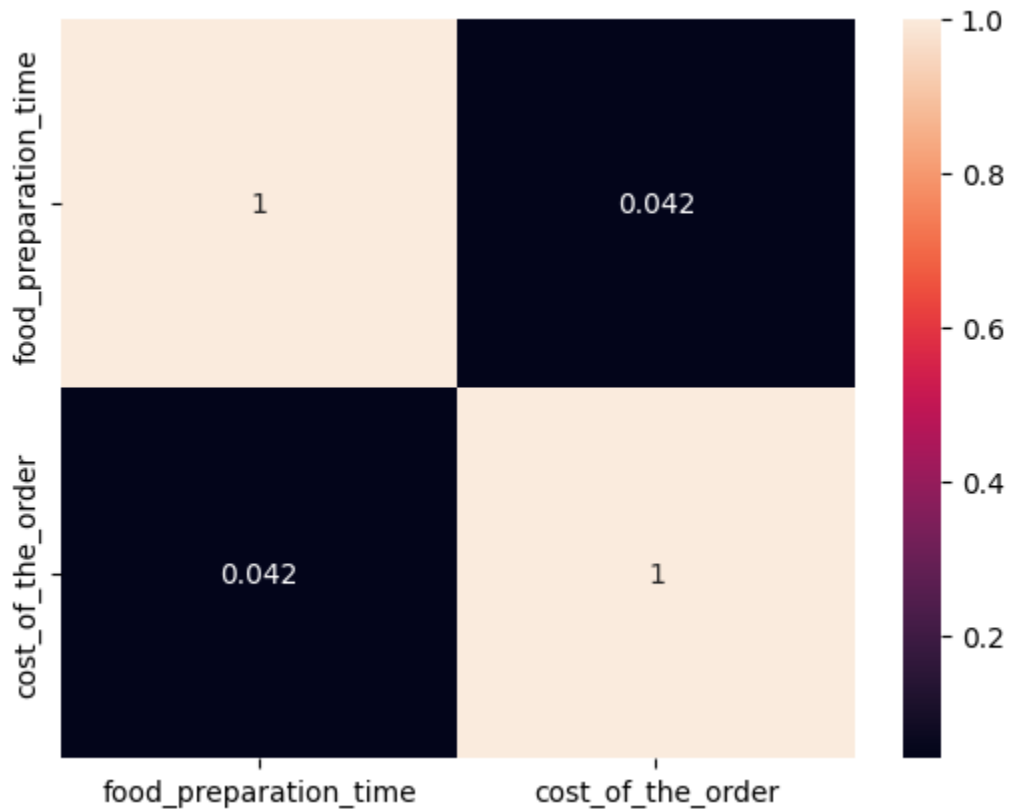
```
In [27]: # No correlation between delivery time and food prep time
sns.heatmap(df[['delivery_time', 'food_preparation_time']].corr(), annot=True)
```

Out[27]: <Axes: >



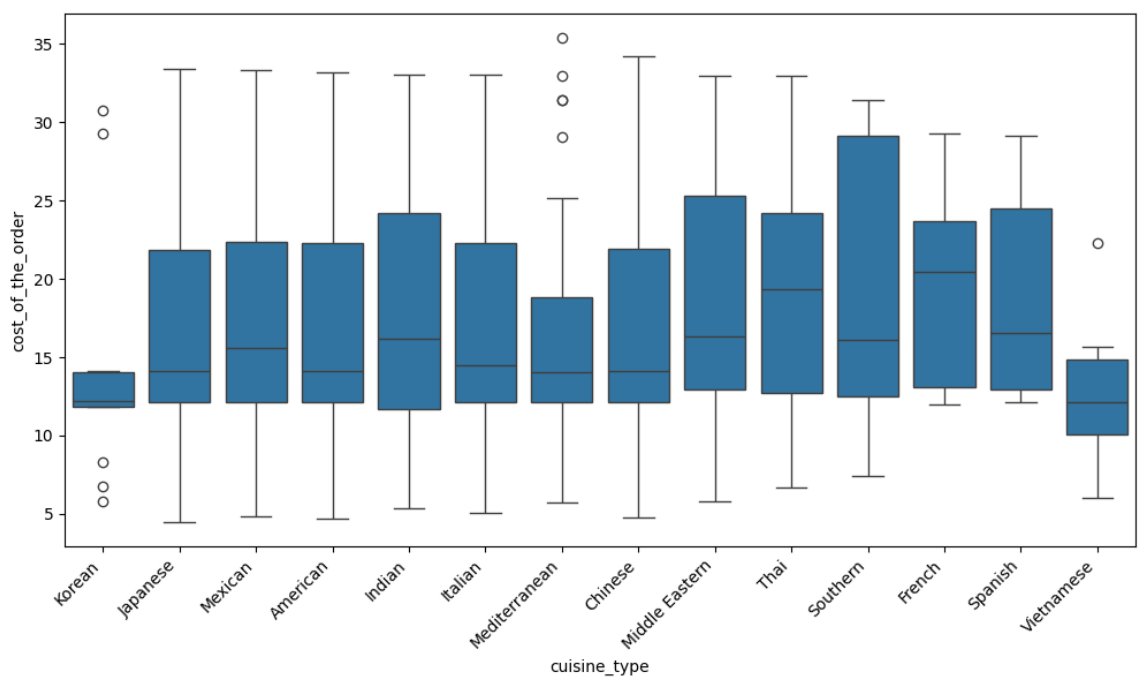
```
In [28]: sns.heatmap(df[['food_preparation_time', 'cost_of_the_order']].corr(),
# No good correlations between these variables
annot=True)
```

Out[28]: <Axes: >



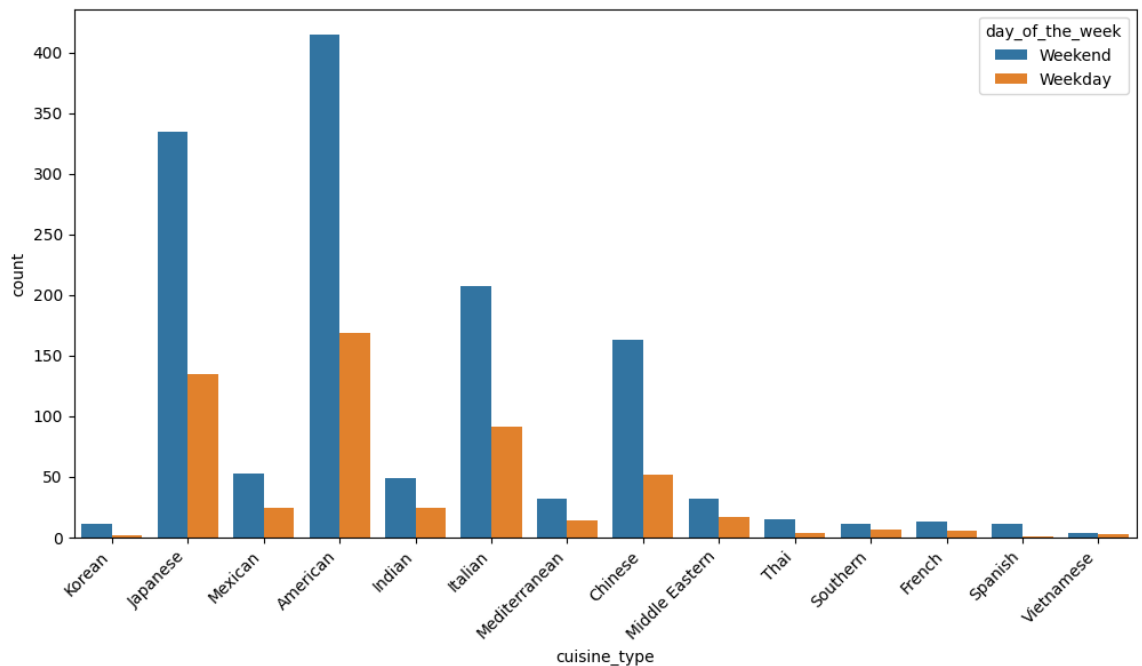
In
[29]:

```
'''Southern cuisine has the highest prices in the Q3
French cuisine has the highest prices in the Q1
Thai has the highest median value at around $20
Mediterranean has the most outliers
Chinese has the highest max and japanese has the lowest min value
'''
plt.figure(figsize=(12, 6))
sns.boxplot(x='cuisine_type', y='cost_of_the_order', data=df)
plt.xticks(rotation=45, ha='right')
plt.show()
```



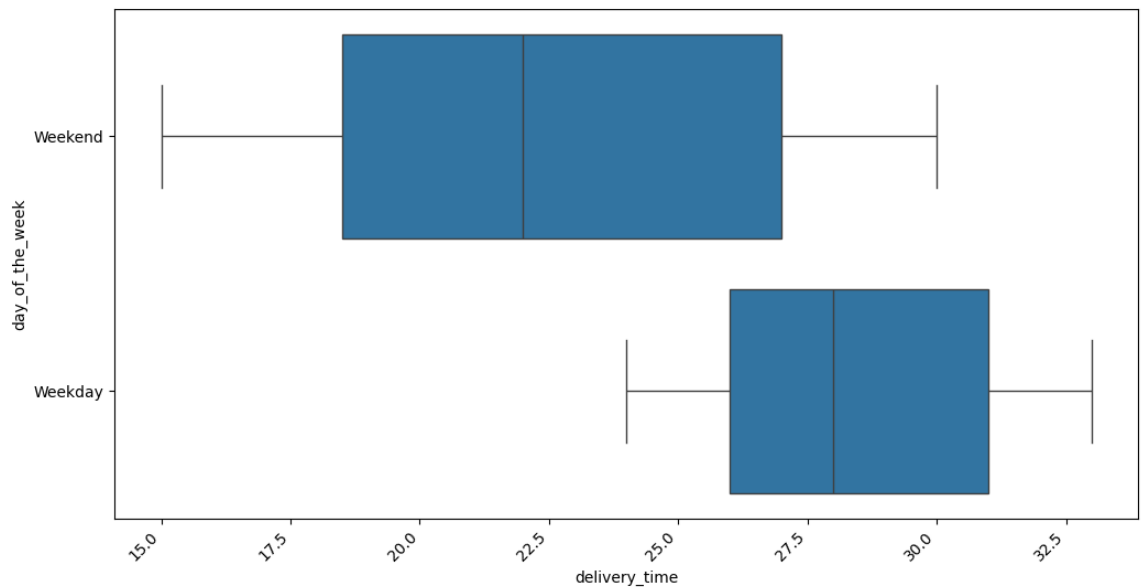
In
[30]:

```
'''  
American has the most orders on both the weekend and weekdays  
Japanese has the second most orders on both the weekend and weekdays  
Italian has the third most orders on both the weekend and weekdays  
All others fall under 200 orders for both weekend and weekdays  
'''  
  
plt.figure(figsize=(12, 6))  
sns.countplot(x='cuisine_type', hue='day_of_the_week', data=df)  
plt.xticks(rotation=45, ha='right')  
plt.show()
```

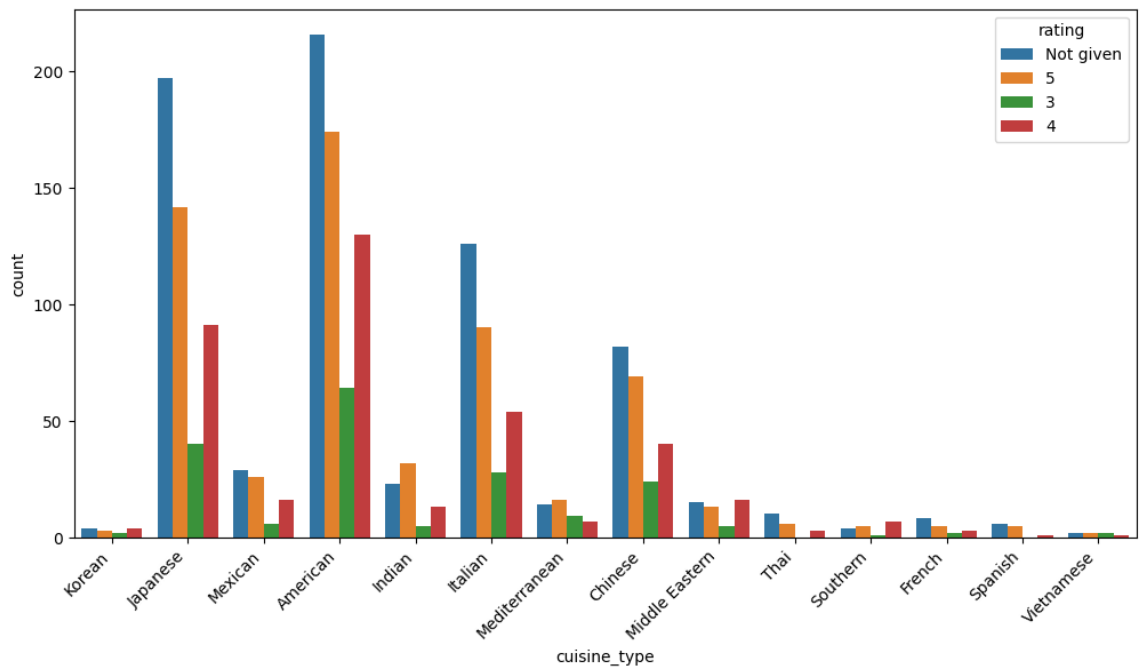


In
[31]:

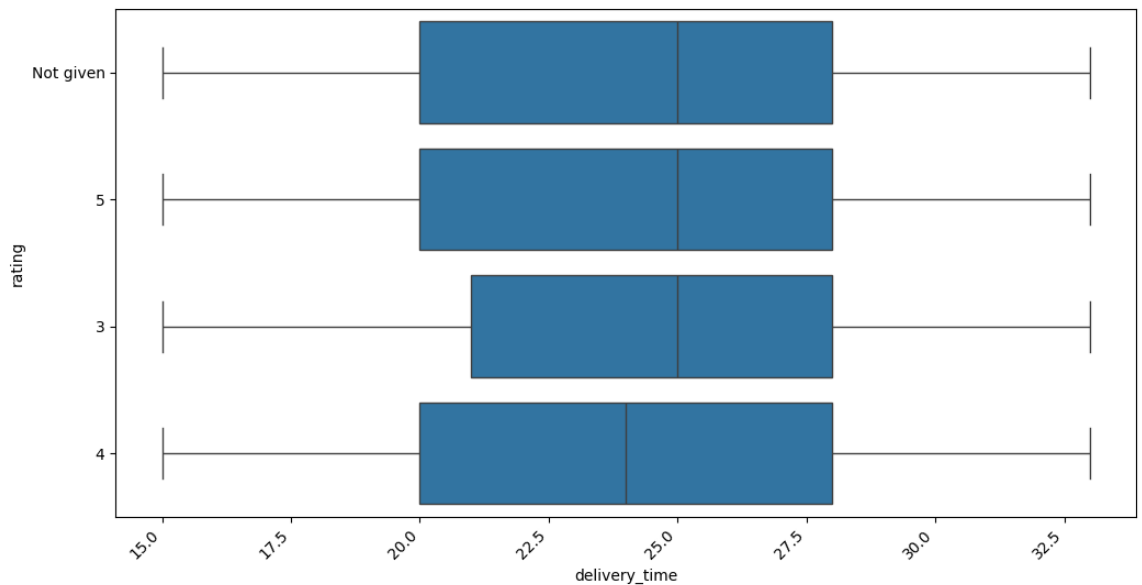
```
# Delivery times on the weekdays are longer, with the median of weekday  
delivery times being more than the Q3 of weekend delivery times  
plt.figure(figsize=(12, 6))  
sns.boxplot(x='delivery_time', y='day_of_the_week', data=df)  
plt.xticks(rotation=45, ha='right')  
plt.show()
```



```
In [32]: # Most of the reviews are not given
# American cuisine has the most reviews overall
# 5 ratings are the most common review other than 'not given'
plt.figure(figsize=(12, 6))
sns.countplot(x='cuisine_type', hue='rating', data=df)
plt.xticks(rotation=45, ha='right')
plt.show()
```



```
In [33]: # The Q1 for 3 ratings is slightly higher than the others
plt.figure(figsize=(12, 6))
sns.boxplot(x='delivery_time', y='rating', data=df)
plt.xticks(rotation=45, ha='right')
plt.show()
```



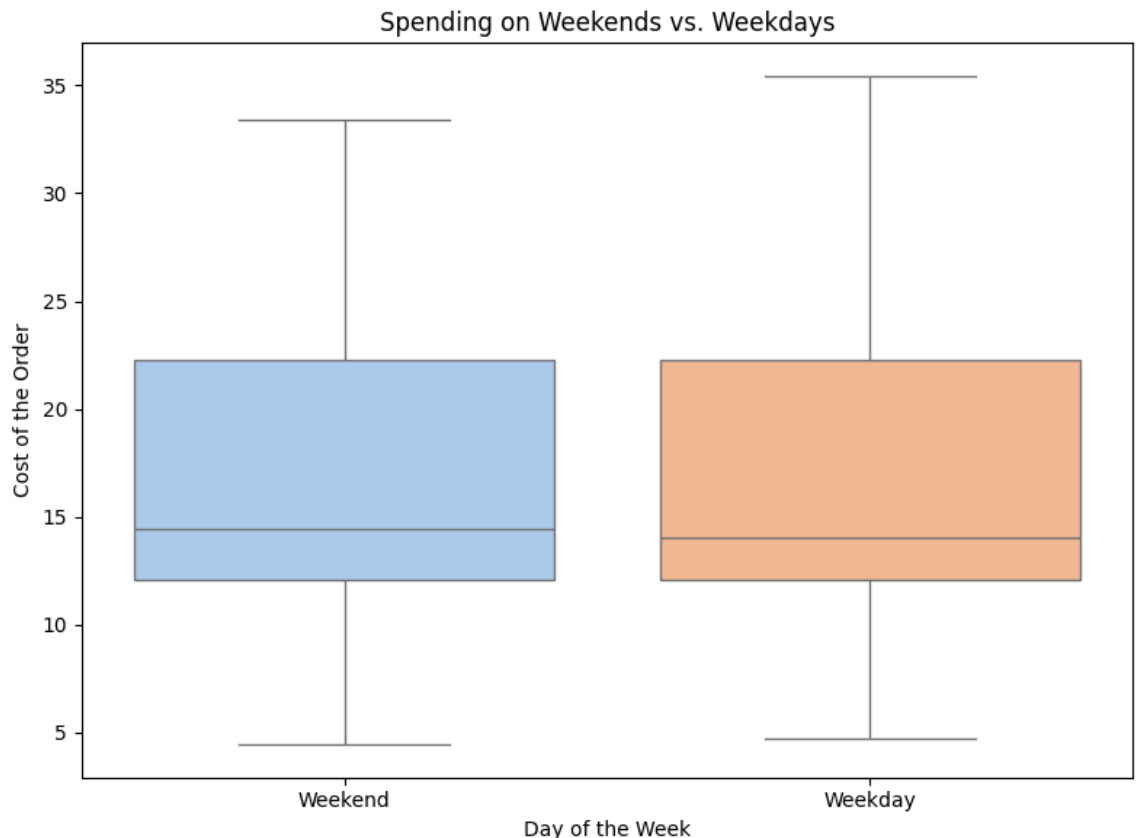
In
[34]:

```
#No correlation between money spent and on the weekdays versus the weekends
plt.figure(figsize=(8, 6))
sns.boxplot(x='day_of_the_week', y='cost_of_the_order', data=df,
palette='pastel')
plt.title('Spending on Weekends vs. Weekdays')
plt.xlabel('Day of the Week')
plt.ylabel('Cost of the Order')
plt.tight_layout()
plt.show()
```

/tmp/ipykernel_11331/3883505632.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.boxplot(x='day_of_the_week', y='cost_of_the_order', data=df,
palette='pastel')
```



Question 13: The company wants to provide a promotional offer in the advertisement of the restaurants. The condition to get the offer is that the restaurants must have a rating count of more than 50 and the average rating should be greater than 4. Find the restaurants fulfilling the criteria to get the promotional offer. [3 marks]

```
In [35]: # Write the code here
df_rating = df[df['rating'] != 'Not given'].copy() # Use .copy() to avoid
SettingWithCopyWarning
df_rating['rating'] = df_rating['rating'].astype(float)

# Calculate rating counts for each restaurant
rating_counts = df_rating['restaurant_name'].value_counts()

# Filter restaurants with more than 50 ratings (r1)
r1 = rating_counts[rating_counts > 50].index

# Calculate average rating for each restaurant (r2)
r2 = df_rating.groupby('restaurant_name')['rating'].mean()

# Filter restaurants with average rating > 4
restaurants_with_high_avg_rating = r2[r2 > 4].index

# Find restaurants that satisfy both conditions
```

```

restaurants_for_promotional_offer = list(set(r1) &
set(restaurants_with_high_avg_rating))

# Display the results
print("Restaurants eligible for the promotional offer (rating count > 50 and
average rating > 4):")
for restaurant in restaurants_for_promotional_offer:
    print(restaurant)

```

```

    Restaurants eligible for the promotional offer (rating count > 50 and
    average rating > 4):
    The Meatball Shop
    Blue Ribbon Fried Chicken
    Blue Ribbon Sushi
    Shake Shack

```

Observations: Shake Shack, Meatball Shop, Blue ribbon fried chicken and sushi all have an average rating greater than 4 and have more than 50 ratings, not a lot of restuarants are qualified

Question 14: The company charges the restaurant 25% on the orders having cost greater than 20 dollars and 15% on the orders having cost greater than 5 dollars. Find the net revenue generated by the company across all orders. [3 marks]

```

In [36]: def calculate_commission(row):
        cost = row['cost_of_the_order']
        if cost > 20:
            return cost * 0.25
        elif cost > 5:
            return cost * 0.15
        else:
            return 0

        df['commission'] = df.apply(calculate_commission, axis=1)
        net_revenue = df['commission'].sum()
        net_revenue

```

Out[36]: np.float64(6166.303)

Observations: The net revenue of the company is \$6166.30

Question 15: The company wants to analyze the total time required to deliver the food. What percentage of orders take more than 60 minutes to get delivered from the time the order is placed? (The food has to be prepared and then delivered.) [2 marks]

```
In [37]: # Write the code here
df_combined = df['delivery_time'] + df['food_preparation_time']
df[df_combined > 60].shape[0] / df.shape[0]
```

Out[37]: 0.1053740779768177

Observations: 10.5% of orders take more than 60 min to be delivered

Question 16: The company wants to analyze the delivery time of the orders on weekdays and weekends. How does the mean delivery time vary during weekdays and weekends? [2 marks]

```
In [38]: # Write the code here
df.groupby('day_of_the_week')['delivery_time'].mean()
```

Out[38]:

	delivery_time
day_of_the_week	
Weekday	28.340037
Weekend	22.470022

dtype: float64

Observations: Weekdays average delivery time is higher than that of the weekends times by around 6 minutes

Conclusion and Business Recommendations

Question 17: What are your conclusions from the analysis? What recommendations would you like to share to help improve the business? (You can use cuisine type and feedback ratings to drive your business recommendations.) [6 marks]

Conclusions

- In conclusion:

- * People order more during the weekends
- * American, Japanese, Italian, and Chinese are the most popular cuisines
- * American food has the most orders throughout the week
- * Weekends delivery times are shorter compared to the weekdays
- * The most popular cuisines all have prices between \$13 and \$25
- * The most popular restaurant is Shake Shack
- * Spending doesn't change from weekends to weekdays
- * Food prep time does not affect any aspect of the data
- * Around a quarter of the ratings were 'not given'

Business Recommendations

- I would recommend to:

- * Focus promotional offers on restaurants like Shake Shack, The Meatball Shop, Blue Ribbon Fried Chicken, and Blue Ribbon Sushi, as they have both high order counts and excellent average ratings (above 4 with over 50 ratings).
- * Since more orders are placed on weekends and delivery times are shorter then, consider running special weekend promotions. Additionally, investigate why weekday delivery times are longer and implement strategies to reduce them.
- * Given the high demand for American, Japanese, Italian, and Chinese cuisines, FoodHub should continue to work with quality restaurants offering these types of food. Highlighting American cuisine specifically, which is most popular throughout the week, can also be a key marketing strategy.
- * With a significant number of orders not being rated, implement incentives to encourage customers to provide feedback.
- * As spending patterns remain similar across weekdays and weekends, a consistent marketing

strategy can be effective. However, emphasize weekend deals to capitalize on the higher order volume during those days.
